

DIY Hydroponic Nutrients

Dragoon Mountains, Arizona

Locally-Sourced Masterblend-Style Formula Guide

PART 1 — Masterblend 4-18-38 Target Profile

The 3-Part System

The Masterblend system uses three separate components mixed in sequence. They must never be pre-combined — mix each into water individually to avoid nutrient lockout.

Component	Amount / 5 gal	Nutrients Provided
Masterblend 4-18-38	12 g	N, P, K, Mg, B, Cu, Fe, Mn, Mo, Zn
Magnesium Sulfate	6 g	Mg, S
Calcium Nitrate	12 g	Ca, N

Target Nutrient Levels (per gallon working solution)

Macronutrient	Target ppm	Micronutrient	Target ppm
Nitrogen (N)	116	Iron (Fe chelated)	2.4
Phosphorus (P)	47	Boron (B)	1.2
Potassium (K)	188	Manganese (Mn)	1.2
Calcium (Ca)	113	Zinc (Zn)	0.3
Magnesium (Mg)	41	Copper (Cu)	0.3
Sulfur (S)	58	Molybdenum (Mo)	0.06

PART 2 — Local Production from the Dragoon Mountains

Part A — Potassium, Phosphorus & Micronutrient Base

(Replaces the 4-18-38 component)

Nutrient	Local Source	Production Method
Potassium (K)	Oak wood ash (abundant in Dragoons)	Burn oak hardwood. Steep ash in water 24 hrs, strain. Use liquid as potash extract.
Phosphorus (P)	Bat guano (Cochise County caves)	Collect guano from bat roosts. Steep in water, aerate 24–48 hrs. Strain before use.
Nitrogen (N)	Aerated guano tea + compost	Aerate guano tea vigorously to encourage nitrifying bacteria to convert $\text{NH}_4^+ \rightarrow \text{NO}_3^-$.
Copper (Cu)	Bisbee copper ore (malachite, azurite)	Dissolve small malachite chip in dilute apple cider vinegar. Dilute heavily before use.
Boron (B)	Desert caliche & Sulphur Springs soil	Boron concentrates in AZ arid soils. Brew concentrated caliche soil tea. Use sparingly.
Iron (Fe chelated)	Red granite clay + oak tannin	Brew strong oak bark/leaf tea. Soak iron-rich red clay in it 48 hrs. Strain and use liquid.

Part B — Calcium Source

(Replaces Calcium Nitrate)

Caliche (calcium carbonate hardpan) is abundant throughout the Sulphur Springs Valley east of the Dragoons. To produce plant-available calcium:

1. Collect caliche chunks from exposed hardpan layers.
2. Dissolve slowly in **apple cider vinegar** (from your fermentation setup) — this produces **calcium acetate**, which is fully plant-available.
3. Combine with nitrified guano tea to provide both calcium and nitrogen together.
4. Dolomite limestone (contains both Ca and Mg) is an alternative if available locally.

Part C — Magnesium + Sulfur

(Replaces Epsom Salt)

The **Sulphur Springs Valley** is named for a reason. Look for **epsomite** — white crystalline mineral deposits around dry springs and mineral seeps. Epsomite IS magnesium sulfate in natural form, identical to Epsom salt. Alternatively, collect sulfur-rich spring water and dissolve dolomite into it.

PART 3 — Micronutrient Production

Nutrient	Local Source	Method
Iron (Fe)	Red granite clay + oak bark	Steep iron-rich clay in oak tannin tea 48 hrs. Tannins chelate the iron into plant-available form.
Manganese (Mn)	Dark metamorphic rocks, streambed sediment	Collect mineral-stained dark streambed mud. Brew as dilute tea. Strain before use.
Zinc (Zn)	Tombstone / Bisbee area zinc-bearing ores	Dissolve zinc ore chip in dilute acid. Heavily dilute before adding to reservoir.
Copper (Cu)	Bisbee copper ores (malachite, azurite)	Dissolve small piece in dilute apple cider vinegar. Tiny amounts only — copper is toxic at high doses.
Boron (B)	Desert caliche, dry lakebeds, borax deposits	AZ desert soils concentrate boron naturally. Sulphur Springs Valley soil tea. Use very sparingly.
Molybdenum (Mo)	Wild legume root nodules (mesquite, clover)	Ferment root nodules from local wild legumes. Provides trace Mo — hardest element to source locally.

PART 4 — Feasibility Summary

Component	Locally Achievable?	Notes
Potassium (K)	✓ Yes	Oak wood ash is an excellent source.
Phosphorus (P)	✓ Yes	Bat guano from Cochise County caves.
Calcium (Ca)	✓ Yes	Caliche is ubiquitous in the area.
Magnesium (Mg)	✓ Likely	Hunt for epsomite deposits in Sulphur Springs Valley.
Sulfur (S)	✓ Yes	Valley named for sulfur mineral deposits.
Iron (Fe) chelated	✓ Yes	Oak tannin + red clay method is effective.
Copper (Cu)	✓ Yes	Bisbee copper ores are world-class.
Boron (B)	✓ Yes	AZ desert soils naturally concentrate boron.
Nitrogen (nitrate)	■ Partial	Hard to get pure nitrate form — aerated guano tea helps.
Mn, Zn (trace)	■ Difficult	Available but hard to dose accurately.
Molybdenum (Mo)	✗ Very hard	Legume tea only provides trace amounts.

Recommended Workflow

Step	Action	Source
1	Brew oak ash potash extract	Local oaks → burn → steep
2	Prepare aerated bat guano tea	Cochise County bat roosts
3	Make caliche-vinegar calcium extract	Sulphur Springs Valley caliche
4	Brew oak tannin + red clay iron chelate	Local granite hillsides
5	Collect epsomite or use purchased Epsom salt	Local mineral seeps OR purchase
6	Make copper micro-dose (Bisbee malachite)	Bisbee area copper ores
7	Brew desert soil boron tea	Sulphur Springs caliche soil
8	Combine all parts separately into reservoir — check EC and pH	Need EC + pH meters

Important: Regardless of how nutrients are sourced, an **EC meter** and **pH meter** are essential. Target EC 1.8–2.4 mS/cm and pH 5.5–6.5 for tomatoes. These instruments cannot be replaced with local materials and should be purchased.